

WORK, POWER & ENERGY

1. When a man pushes a wall he gets tired because of:
 - (1) Internal energy utilised in contracting and extending muscles
 - (2) By doing positive work on wall
 - (3) By doing negative work on wall
 - (4) None
2. When a man starts running from rest, then work done by friction is:
 - (1) zero
 - (2) positive
 - (3) negative
 - (4) None of these
3. Work done by centripetal force on a particle performing circular motion is:
 - (1) Zero
 - (2) Positive
 - (3) Negative
 - (4) Can't be said
4. Work done by conservative force is:
 - (1) ΔK
 - (2) ΔU
 - (3) $-\Delta U$
 - (4) $\Delta K + \Delta U$
5. Work done by kinetic friction is:
 - (1) (+)ve
 - (2) (-)ve
 - (3) 0
 - (4) May be of any sign.
6. If mechanical energy is constant, then speed of particle:
 - (1) Increases continuously
 - (2) decreases continuously
 - (3) will be constant
 - (4) can't be said
7. If a block of mass M slides on a rough table by a horizontal string:
 - (1) Work done by tension is equal to change in K.E.
 - (2) Work done by friction is equal to change in K.E.
 - (3) Work done by friction converts to dissipative losses
 - (4) Work done by tension is equal to change in P.E.
8. Gravitational potential energy of two mass system at any given instant does not depend on :
 - (1) Their separation
 - (2) Gravitational force between them
 - (3) Value of masses
 - (4) Velocities of masses
9. Work done by air resistance on a body falling freely is :
 - (1) (+)ve
 - (2) (-)ve
 - (3) 0
 - (4) Can be anything
10. Choose incorrect statement:
 - (1) Energy stored in spring is same when it is stretched or compressed by same length
 - (2) Power of centripetal force is always zero in circular motion
 - (3) During inelastic collision of two masses K.E. lost during collision recovers partially
 - (4) Work done by gravity in projectile motion is always positive
11. Choose correct statement:
 - (1) Slope of force-displacement graph is work done
 - (2) Area of P.E.-position graph is force
 - (3) Slope of work-time graph is power
 - (4) Area of acceleration-position graph is work-done
12. The total work done on a particle is equal to the change in its kinetic energy:
 - (1) Always
 - (2) Only if the forces acting on it are conservative
 - (3) Only if gravitational force acts on it
 - (4) Only if elastic force acts on it
13. A power (P) versus position (x) graph for a particle is shown below. The particle is of mass $\frac{10}{7}$ kg and is moving in the positive x direction. Its initial position is $x = 0$ and initial velocity is 1 m/s. The velocity at $x = 10$ m is :

 - (1) 4 m/s
 - (2) 2 m/s
 - (3) $3\sqrt{2}$ m/s
 - (4) $\frac{100}{3}$ m/s

14. Which of the following statement is correct?
- (1) When a conservative force does positive work on a body, then potential energy of body increases
 - (2) A non-conservative force always decreases mechanical energy of the system
 - (3) Total work done by friction force on a complete system can never be positive
 - (4) Internal forces can never change the kinetic energy of the system
15. Which of the following statements is incorrect?
- (1) If momentum of an object is zero its kinetic energy must be zero
 - (2) If kinetic energy of an object is zero then its momentum must be zero
 - (3) If momentum of a system is zero, then its kinetic energy must be zero
 - (4) If kinetic energy of a system is zero, then its momentum must be zero

16. Which statement is correct?
- (1) In an elastic collision of two balls, the total kinetic energy is conserved during the short time of collision of balls
 - (2) In an elastic collision of two balls, the total momentum is conserved during the short time of collision of balls
 - (3) Both are correct
 - (4) Both of incorrect